

Week : 1

Day 1: Introduction to Mobile App Development

Date: 27/05/2025

1. Introduction to Android

- Android is an updated version of the Linux operating system.
- It was introduced in **2008**.
- Officially introduced as a subject in **2015 (GTU Curriculum)**.
- Developed by **Google** under the **Open Handset Alliance (OHA)**.
- OHA is a consortium led by Google that includes various companies to support Android development.

2. Supported Languages in Android Development

- **Java**
 - Tools: JDK, SDK (Software Development Kit), Eclipse (IDE)
- **Kotlin**

3. Android Development Environment Setup

- Supported Programming Languages: Java and Kotlin
- Required Tools:
 - **JDK (Java Development Kit)**
 - **SDK (Software Development Kit)**
 - **IDE (Eclipse/Android Studio)**

4. Android Evolution Flow Diagram

IOT (Internet of Things)



IOE (Internet of Everything)



Linux Updated Version (2008)



Open Handset Alliance (OHA) → Google → Android



Handover Supports
by Google Java & Kotlin

Languages Used in Android:

Java → SDK, Eclipse → Android Development Environment Setup

Kotlin → Android Development Environment Setup

5. Android Marketplace

- Known as **Google Play Store**
- Acts as a **Market Place (Growth Engine)** for distributing Android apps.

6. Android Studio

- **Latest version:** Android Studio 16

7. Virtual Machines in Android

- **JVM (Java Virtual Machine):** Used in Java
- **DVM (Dalvik Virtual Machine):** Used in Android
 - DVM is the **compiler** for Android that executes bytecode and runs the app.

Day 2: Android Project Framework

Date: 28/05/2025

1. **Gradle** is used in Android Studio for compiling and deploying Android projects.

2. Gradle can **compile** and **deploy** your project to a **mobile** device or a **laptop**.
3. It is a **build tool** responsible for automation of build tasks.
4. The **process** involves:
 - Writing code in Android Studio (on laptop)
 - Gradle compiles the code
 - Gradle deploys the build to a mobile device for testing or use

Diagram Analysis:

- Shows a flow:
 - Gradle → compile & deploy →
 - Targets: mobile and laptop
- Gujarati labels explain that the build is done on the laptop and deployment is done on the mobile.

Android Project Folder Structure

1. Project Folders:

- manifests: Contains AndroidManifest.xml (declares components, permissions, etc.)
- java: Contains all Java/Kotlin source code
- res: Resources folder with subfolders:
 - drawable: For images, shapes, and vectors
 - layout: XML files for UI design
 - mipmap: App launcher icons
 - values: Strings, colors, styles, etc.

2. Gradle Scripts:

- Automates build tasks
- Compiles code, packages APK

3. Important Notes:

- Gradle build happens in Android Studio
- Deployment (running the app) happens on a mobile device

Image Rules & Android Widgets

1. Image Handling:

- Place all images in drawable folder
- File names must use only **uppercase or lowercase letters**
- **No symbols or digits** in file names

2. Widgets Overview:

Widget Type	Examples
Text widgets	TextView, EditText
Button widgets	Button, ToggleButton
Layout widgets	LinearLayout, RelativeLayout
Image & Media widgets	ImageView, VideoView

Progress & Rating widgets	ProgressBar, SeekBar
Picker widgets	DatePicker, TimePicker
List & Collection widgets	ListView, Spinner
Web & Search widgets	WebView, SearchView
App widgets	ClockWidget, WeatherWidget

Day 3: Installation of Android Studio

Date: 29/05/2025

Step-by-Step Installation

1. Download Android Studio

- Go to the official site: <https://developer.android.com/studio>
- Click on “**Download Android Studio**”.
- Accept the terms and download the .exe file (around 1 GB).

2. Run the Installer

- Double-click the downloaded .exe file.
- Click **Next** on the setup screen.

3. Choose Components

- Make sure these are selected:
 - Android Studio
 - Android Virtual Device (AVD)
- Click **Next**.

4. Choose Installation Path

- Leave the default path or change it if needed (e.g., C:\Program Files\Android\Android Studio).
- Click **Next** → **Install**.

5. Wait for Installation to Complete

- It will take a few minutes.
- Once done, click **Finish**.

6. Launch Android Studio

- After finishing, Android Studio will open.
- On first launch, it will ask for setup.

7. Setup Wizard

- Select **Standard Installation**.
- Click **Next** → It will download SDK, Emulator, and other tools.
- Wait for components to download and install (requires internet).

8. Select Theme

- Choose your preferred theme (Light or Dark).
- Click **Next** → **Finish**.

9. Start a New Project

- After setup, you will see the Android Studio welcome screen.

- Click “**New Project**” to begin coding.

Notes:

- Make sure your system has:
 - Minimum **8 GB RAM** (recommended)
 - **64-bit Windows 10/11**
 - At least **4 GB of free disk space**
- Keep internet connection active during setup for SDK downloads.

Day: 4 Intent Filters and App Components

Date: 30/05/2025

1. **Intent filter:** It is defined in the manifest folder. It is used for **Activity-to-Activity navigation**.
2. In Android, **Intent** is a mechanism that tells the Operating System which type of intent an app component can handle.
3. App components include: **Activity, Service, BroadcastReceiver**, and so on.
4. There are **two ways to implement intent filters**:
 1. **Implicit:** Activity to another framework component (like Google Maps or Browser).
 2. **Explicit:** Activity to another activity within the same application.
5. **Intent filters** help Android know which activity should respond to a given intent.
6. **Implicit Intents** do not name a specific component — the system decides the best match.
7. **Explicit Intents** directly specify the target component using the class name.
8. Intents can also be used to pass **data between activities** using `putExtra()` and `getExtra()` methods.
9. The `<intent-filter>` tag is used inside the `<activity>` tag in the `AndroidManifest.xml` file.

Day : 5 Android Project: "Sum of Two Numbers App"

Date: 31/05/2025

Overview:

This Android application allows users to enter two numbers and calculates their sum. The app interface includes:

- Two input fields (EditText) labeled `Number1` and `Number2` for entering values.
- A button labeled `ADD`, which when clicked, performs the addition.
- A display area (could be a `TextView` or `Toast/Result` display) that shows the result of the addition (e.g., `15.0` in your example).

The app title shown at the top is `My Application1`, but logically it represents the purpose: `Sum of Two Numbers`.

Functionality:

- The user enters the first number in the `Number1` field.
- The user enters the second number in the `Number2` field.
- On pressing the `ADD` button, the app:
 - Reads both numbers.
 - Converts them to numeric values (if they are strings).
 - Calculates the sum.
 - Displays the result (for example, `10 + 5 = 15.0`).

XML Layout Code (activity_main.xml)

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <TextView
```



```
android:id="@+id/txt1"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:text="Number1"
app:layout_constraintBottom_toBottomOf="parent"
app:layout_constraintEnd_toEndOf="parent"
app:layout_constraintHorizontal_bias="0.147"
app:layout_constraintStart_toStartOf="parent"
app:layout_constraintTop_toTopOf="parent"
app:layout_constraintVertical_bias="0.165" />
```

<TextView

```
android:id="@+id/txt2"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:text="Number2"
app:layout_constraintBottom_toBottomOf="parent"
app:layout_constraintEnd_toEndOf="parent"
app:layout_constraintHorizontal_bias="0.147"
app:layout_constraintStart_toStartOf="parent"
app:layout_constraintTop_toTopOf="parent"
app:layout_constraintVertical_bias="0.299" />
```

<EditText

```
android:id="@+id/plt1"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:ems="10"
android:inputType="textPersonName"
android:text=""
app:layout_constraintBottom_toBottomOf="parent"
app:layout_constraintEnd_toEndOf="parent"
app:layout_constraintHorizontal_bias="0.791"
```

```
app:layout_constraintStart_toStartOf="parent"  
app:layout_constraintTop_toTopOf="parent"  
app:layout_constraintVertical_bias="0.17" />
```

<EditText

```
android:id="@+id/plt2"  
android:layout_width="wrap_content"  
android:layout_height="wrap_content"  
android:ems="10"  
android:inputType="textPersonName"  
android:text=""  
app:layout_constraintBottom_toBottomOf="parent"  
app:layout_constraintEnd_toEndOf="parent"  
app:layout_constraintHorizontal_bias="0.796"  
app:layout_constraintStart_toStartOf="parent"  
app:layout_constraintTop_toTopOf="parent"  
app:layout_constraintVertical_bias="0.31" />
```

<Button

```
android:id="@+id/btna"  
android:layout_width="wrap_content"  
android:layout_height="wrap_content"  
android:text="ADD"  
app:layout_constraintBottom_toBottomOf="parent"  
app:layout_constraintEnd_toEndOf="parent"  
app:layout_constraintStart_toStartOf="parent"  
app:layout_constraintTop_toTopOf="parent"  
android:onClick="total" />
```

</androidx.constraintlayout.widget.ConstraintLayout>

Java Code (MainActivity.java)

```

package com.example.myapplication1;

import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;
import android.widget.Toast;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }

    public void total(View v) {
        Button b = findViewById(R.id.btna);
        EditText e1 = findViewById(R.id.plt1);
        EditText e2 = findViewById(R.id.plt2);
        String n1 = e1.getText().toString();
        String n2 = e2.getText().toString();
        Double result = Double.parseDouble(n1) +
Double.parseDouble(n2);
        String sum = String.valueOf(result);
        Toast.makeText(this, sum, Toast.LENGTH_LONG).show();
    }
}

```

Day : 6 Android Project: "Subtracting Two Numbers"

Date: 02/06/2025

Overview:

This Android application helps users subtract two numbers easily. The interface contains:

- Two input fields (EditText) labeled as Num 1 and Num 2, where users enter numbers.
- A button labeled SUB that performs the subtraction when clicked.
- A result display at the bottom showing the calculated difference (e.g., $10 - 5 = 5.0$).

The app title shows as My Application2, but its purpose is to perform subtraction of two numbers.

Functionality:

1. The user enters the first number (e.g., 10) in the Num 1 field.
2. The user enters the second number (e.g., 5) in the Num 2 field.
3. On clicking the SUB button:
 - a. The app reads both numbers.
 - b. Converts them into numeric format (if entered as text).
 - c. Subtracts Num 2 from Num 1.
 - d. Displays the result (5.0 in this example).

XML Layout Code (activity_main.xml)

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <TextView
```

```
android:id="@+id/txt1"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:text="Num 1:"
app:layout_constraintBottom_toBottomOf="parent"
app:layout_constraintEnd_toEndOf="parent"
app:layout_constraintHorizontal_bias="0.152"
app:layout_constraintStart_toStartOf="parent"
app:layout_constraintTop_toTopOf="parent"
app:layout_constraintVertical_bias="0.148" />
```

<TextView

```
android:id="@+id/txt2"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:text="Num 2:"
app:layout_constraintBottom_toBottomOf="parent"
app:layout_constraintEnd_toEndOf="parent"
app:layout_constraintHorizontal_bias="0.152"
app:layout_constraintStart_toStartOf="parent"
app:layout_constraintTop_toTopOf="parent"
app:layout_constraintVertical_bias="0.283" />
```

<EditText

```
android:id="@+id/plt1"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:ems="10"
android:inputType="textPersonName"
android:text=""
app:layout_constraintBottom_toBottomOf="parent"
app:layout_constraintEnd_toEndOf="parent"
app:layout_constraintHorizontal_bias="0.701"
app:layout_constraintStart_toStartOf="parent"
app:layout_constraintTop_toTopOf="parent"
```

```
app:layout_constraintVertical_bias="0.154" />
```

```
<EditText
```

```
    android:id="@+id/plt2"  
    android:layout_width="wrap_content"  
    android:layout_height="wrap_content"  
    android:ems="10"  
    android:inputType="textPersonName"  
    android:text=""  
    app:layout_constraintBottom_toBottomOf="parent"  
    app:layout_constraintEnd_toEndOf="parent"  
    app:layout_constraintHorizontal_bias="0.706"  
    app:layout_constraintStart_toStartOf="parent"  
    app:layout_constraintTop_toTopOf="parent"  
    app:layout_constraintVertical_bias="0.294" />
```

```
<Button
```

```
    android:id="@+id/btnA"  
    android:layout_width="wrap_content"  
    android:layout_height="wrap_content"  
    android:text="SUB"  
    app:layout_constraintBottom_toBottomOf="parent"  
    app:layout_constraintEnd_toEndOf="parent"  
    app:layout_constraintHorizontal_bias="0.498"  
    app:layout_constraintStart_toStartOf="parent"  
    app:layout_constraintTop_toTopOf="parent"  
    app:layout_constraintVertical_bias="0.707"  
    android:onClick="sub" />
```

```
</androidx.constraintlayout.widget.ConstraintLayout>
```

Java Code (MainActivity.java)

```
package com.example.myapplication2;
```

```
import androidx.appcompat.app.AppCompatActivity;
```

```

import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }
    public void sub(View v){
        Button b = findViewById(R.id.btnA);
        EditText E1 = findViewById(R.id.plt1);
        EditText E2 = findViewById(R.id.plt2);
        String num1 = E1.getText().toString();
        String num2 = E2.getText().toString();
        Double r =Double.parseDouble(num1)-
Double.parseDouble(num2);
        String sub = String.valueOf(r);
        Toast.makeText(this,sub,Toast.LENGTH_LONG).show();

    }
}

```

Week : 2

Day : 1 Android Project:

**" User Interaction with Buttons to Display Snackbar Messages" OR
 "Basic Android Calculator Application for Arithmetic Operations"**

Date: 03/06/2025

Overview:

Your Android app titled "My Application3" presents a simple, clean UI with:

- A black background (dark theme style)
- A header: My Application3
- Three vertically arranged buttons labeled CLICK HERE with a nice purple background
- A floating button at the bottom that dynamically changes text to show:
 - Hello
 - Good Morning
 - Namaste

Each button appears to perform an action that updates the floating button text depending on which button is clicked.

Functionality:

- First button (CLICK HERE) → When clicked, floating button shows Hello
- Second button (CLICK HERE) → When clicked, floating button shows Good Morning
- Third button (CLICK HERE) → When clicked, floating button shows Namaste
- This shows that:
 - Buttons trigger changes on another component (floating button)
 - App demonstrates use of `setOnClickListener` to handle button clicks
 - Shows how to dynamically update UI elements at runtime

XML Layout Code (activity_main.xml)

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
```



```
android:layout_height="match_parent"
tools:context=".MainActivity">
```

<Button

```
    android:id="@+id/btnh"
    android:layout_width="176dp"
    android:layout_height="69dp"
    android:text="Click here"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.447"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintVertical_bias="0.124"
    android:onClick="Hello" />
```

<Button

```
    android:id="@+id/btng"
    android:layout_width="169dp"
    android:layout_height="64dp"
    android:text="Click here"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.447"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintVertical_bias="0.37"
    android:onClick="Good" />
```

<Button

```
    android:id="@+id/btnn"
    android:layout_width="187dp"
    android:layout_height="60dp"
```

```
    android:text="Click here"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.447"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintVertical_bias="0.623"
    android:onClick="Num"/>
```

```
</androidx.constraintlayout.widget.ConstraintLayout>
```

Java Code (MainActivity.java)

```
package com.example.myapplication3;

import androidx.appcompat.app.AppCompatActivity;

import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.Toast;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }

    public void Hello(View v) {
        Toast.makeText(this, "Hello", Toast.LENGTH_LONG).show();
    }

    public void Good(View v) {
        Toast.makeText(this, "Good Morning",
```

```
Toast.LENGTH_LONG).show();
    }
    public void Num(View v){
        Toast.makeText(this,"Namaste",Toast.LENGTH_LONG).show();
    }
}
```

Android Project: "Basic Android Calculator Application for Arithmetic Operations"

Overview:

1. App Title: Displayed as “My Application4” – likely the default name from Android Studio when the app was created.
2. Input Fields: Number 1 and Number 2: Text input fields where users enter numerical values.
3. Operation Buttons: +, -, *, / buttons represent basic arithmetic operations (addition, subtraction, multiplication, division).
4. Result Display: After pressing an operation, the result (here 48.0) is shown in a text area at the bottom.

Functionality:

- The app takes two numbers from the user.
- When a button (like *) is pressed, it calculates and displays the result.
- Example from screenshot:
 - Input: 12 and 4
 - Pressed: *
 - Output: 48.0 (as expected)

XML Layout Code (activity_main.xml)

```
<?xml version="1.0" encoding="utf-8"?>
```

```
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">
```

```
<TextView
    android:id="@+id/txt1"
    android:layout_width="104dp"
    android:layout_height="40dp"
    android:text="Number 1"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.135"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintVertical_bias="0.113" />
```

```
<TextView
    android:id="@+id/txt2"
    android:layout_width="105dp"
    android:layout_height="40dp"
    android:text="Number 2"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.133"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintVertical_bias="0.235" />
```

```
<EditText
```

```
android:id="@+id/plt1"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:ems="10"
android:inputType="textPersonName"
android:text=""
app:layout_constraintBottom_toBottomOf="parent"
app:layout_constraintEnd_toEndOf="parent"
app:layout_constraintHorizontal_bias="0.805"
app:layout_constraintStart_toStartOf="parent"
app:layout_constraintTop_toTopOf="parent"
app:layout_constraintVertical_bias="0.106" />
```

<EditText

```
android:id="@+id/plt2"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:ems="10"
android:inputType="textPersonName"
android:text=""
app:layout_constraintBottom_toBottomOf="parent"
app:layout_constraintEnd_toEndOf="parent"
app:layout_constraintHorizontal_bias="0.81"
app:layout_constraintStart_toStartOf="parent"
app:layout_constraintTop_toTopOf="parent"
app:layout_constraintVertical_bias="0.23" />
```

<Button

```
android:id="@+id/btnp"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:text="+"
app:layout_constraintBottom_toBottomOf="parent"
```

```
app:layout_constraintEnd_toEndOf="parent"
app:layout_constraintHorizontal_bias="0.164"
app:layout_constraintStart_toStartOf="parent"
app:layout_constraintTop_toTopOf="parent"
app:layout_constraintVertical_bias="0.543" />
```

<Button

```
android:id="@+id/btns"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:text="-"
app:layout_constraintBottom_toBottomOf="parent"
app:layout_constraintEnd_toEndOf="parent"
app:layout_constraintHorizontal_bias="0.703"
app:layout_constraintStart_toStartOf="parent"
app:layout_constraintTop_toTopOf="parent"
app:layout_constraintVertical_bias="0.543" />
```

<Button

```
android:id="@+id/btnm"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:text="*"
app:layout_constraintBottom_toBottomOf="parent"
app:layout_constraintEnd_toEndOf="parent"
app:layout_constraintHorizontal_bias="0.164"
app:layout_constraintStart_toStartOf="parent"
app:layout_constraintTop_toTopOf="parent"
app:layout_constraintVertical_bias="0.72" />
```

<Button

```
android:id="@+id/btnd"
android:layout_width="wrap_content"
```

```
android:layout_height="wrap_content"
android:text="/"
app:layout_constraintBottom_toBottomOf="parent"
app:layout_constraintEnd_toEndOf="parent"
app:layout_constraintHorizontal_bias="0.703"
app:layout_constraintStart_toStartOf="parent"
app:layout_constraintTop_toTopOf="parent"
app:layout_constraintVertical_bias="0.72" />
```

<EditText

```
android:id="@+id/plt3"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:ems="10"
android:inputType="textPersonName"
android:text=""
app:layout_constraintBottom_toBottomOf="parent"
app:layout_constraintEnd_toEndOf="parent"
app:layout_constraintHorizontal_bias="0.497"
app:layout_constraintStart_toStartOf="parent"
app:layout_constraintTop_toTopOf="parent"
app:layout_constraintVertical_bias="0.905" />
```

</androidx.constraintlayout.widget.ConstraintLayout>

Java Code (MainActivity.java)

```
package com.example.myapplication4;

import androidx.appcompat.app.AppCompatActivity;

import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
```

```
public class MainActivity extends AppCompatActivity {
    EditText num1,num2,result;
    Button addbtn,subbtn,mulbtn,divbtn;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        num1 = findViewById(R.id.plt1);
        num2 = findViewById(R.id.plt2);
        result = findViewById(R.id.plt3);

        addbtn = findViewById(R.id.btnp);
        subbtn = findViewById(R.id.btns);
        mulbtn = findViewById(R.id.btnm);
        divbtn = findViewById(R.id.btnd);

        addbtn.setOnClickListener(v→calculate('+'));
        subbtn.setOnClickListener(v→calculate('-'));
        mulbtn.setOnClickListener(v→calculate('*'));
        divbtn.setOnClickListener(v→calculate('/'));
    }

    private void calculate(char operator){
        String input1 = num1.getText().toString();
        String input2 = num2.getText().toString();

        if(!input1.isEmpty()&&!input2.isEmpty()){
            double a = Double.parseDouble(input1);
            double b = Double.parseDouble(input2);
            double res = 0;
```



```

        switch (operator){
            case '+':res = a + b;break;
            case '-':res = a - b;break;
            case '*':res = a * b;break;
            case '/':
                if (b!=0)res = a / b;
                else {
                    result.setText("Can not divide by 0");
                    return;
                }
            break;
        }
        result.setText(String.valueOf(res));
    }else {
        result.setText("Enter numbers");
    }
}
}
}

```

Day : 2 Android Project:

"Android App Functionality: Navigating to Web Content via Button Click"OR" Android App: External URL Integration with Image Buttons"

Date: 04/06/2025

Overview:

This app has a Button labeled “CLICK HERE”. When clicked, it launches the Google.com website in the device’s default browser. The second screenshot confirms the successful redirection to Google.

Functionality:

- UI: Button

- Uses an **Intent** to open a URL in a browser

XML Layout Code (activity_main.xml)

```
<?xml version="1.0" encoding="utf-8"?>
<android.support.constraint.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <Button
        android:id="@+id/btnc"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Click here"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        android:onClick="Click" />

</android.support.constraint.ConstraintLayout>
```

Java Code (MainActivity.java)

```
package com.example.myapplication6;

import android.content.Intent;
import android.net.Uri;
import android.support.v7.app.AppCompatActivity;
import android.os.Bundle;
```

```

import android.view.View;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }

    public void Click(View v) {
        Intent i = new Intent(android.content.Intent.ACTION_VIEW,
Uri.parse("http://www.google.co.in"));
        startActivity(i);
    }
}

```

Android Project: "External URL Integration with Image Buttons"

Overview:

This android app named “My Application7” with three ImageButton, each display a logo:

1. Google Logo
2. Instagram Logo
3. Facebook Logo

Functionality:

- When the user taps on an image:
 - **Google logo** → opens <https://www.google.com>
 - **Instagram logo** → opens <https://www.instagram.com>
 - **Facebook logo** → opens <https://www.facebook.com>

XML Layout Code (activity_main.xml)

```
<?xml version="1.0" encoding="utf-8"?>
<android.support.constraint.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <ImageButton
        android:id="@+id/btng"
        android:layout_width="326dp"
        android:layout_height="203dp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.494"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_bias="0.077"
        app:srcCompat="@mipmap/google"
        android:onClick="google"/>

    <ImageButton
        android:id="@+id/btni"
        android:layout_width="333dp"
        android:layout_height="190dp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.538"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
```

```
app:layout_constraintVertical_bias="0.499"  
app:srcCompat="@mipmap/insta"  
android:onClick="insta" />
```

```
<ImageButton  
    android:id="@+id/btnf"  
    android:layout_width="329dp"  
    android:layout_height="183dp"  
    app:layout_constraintBottom_toBottomOf="parent"  
    app:layout_constraintEnd_toEndOf="parent"  
    app:layout_constraintHorizontal_bias="0.524"  
    app:layout_constraintStart_toStartOf="parent"  
    app:layout_constraintTop_toTopOf="parent"  
    app:layout_constraintVertical_bias="0.883"  
    app:srcCompat="@mipmap/facebook"  
    android:onClick="face" />
```

```
</android.support.constraint.ConstraintLayout>
```

Java Code (MainActivity.java)

```
package com.example.myapplication7;  
  
import android.content.Intent;  
import android.net.Uri;  
import android.support.v7.app.AppCompatActivity;  
import android.os.Bundle;  
import android.view.View;  
  
public class MainActivity extends AppCompatActivity {  
  
    @Override  
    protected void onCreate(Bundle savedInstanceState) {  
        super.onCreate(savedInstanceState);  
    }  
}
```

```

        setContentView(R.layout.activity_main);
    }
    public void google(View v){
        Intent i = new Intent(android.content.Intent.ACTION_VIEW,
Uri.parse("http://www.google.co.in"));
        startActivity(i);
    }
    public void insta(View v){
        Intent j = new Intent(android.content.Intent.ACTION_VIEW,
Uri.parse("https://www.instagram.com"));
        startActivity(j);
    }
    public void face(View v){
        Intent k = new Intent(android.content.Intent.ACTION_VIEW,
Uri.parse("https://www.facebook.com"));
        startActivity(k);
    }
}

```

Day : 3 Android Project:

"Interest Selection Application" OR "Menu Selection App"

Date: 05/06/2025

Overview:

1. Title Bar: App name is displayed as “My Application8” with a purple background, which gives it a colorful look.
2. Heading: "Hobbies" is shown as a label to guide users.
3. CheckBoxes: You have 4 hobbies listed:
 - Reading
 - Singing

- Dancing
- Playing

The checkboxes let users select multiple hobbies at once.

4. Show Button: When the “SHOW” button is pressed, it captures the selected hobbies and displays them.
5. Toast Message: At the bottom:
"You are selected: ReadingSinging"
This tells the user what options they picked.

Functionality:

- The user selects one or more hobbies by checking the boxes.
- On clicking SHOW, the app:
- Checks the state (`isChecked`) of each `CheckBox`.
- Appends selected hobbies into a string.
 - Displays the string using a Toast message.

XML Layout Code (activity_main.xml)

```
<?xml version="1.0" encoding="utf-8"?>
<android.support.constraint.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <CheckBox
        android:id="@+id/btnc"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
```

```
android:text="Reading"
app:layout_constraintBottom_toBottomOf="parent"
app:layout_constraintEnd_toEndOf="parent"
app:layout_constraintHorizontal_bias="0.23"
app:layout_constraintStart_toStartOf="parent"
app:layout_constraintTop_toTopOf="parent"
app:layout_constraintVertical_bias="0.223" />
```

<CheckBox

```
android:id="@+id/btnc1"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:text="Singing"
app:layout_constraintBottom_toBottomOf="parent"
app:layout_constraintEnd_toEndOf="parent"
app:layout_constraintHorizontal_bias="0.23"
app:layout_constraintStart_toStartOf="parent"
app:layout_constraintTop_toTopOf="parent"
app:layout_constraintVertical_bias="0.324" />
```

<CheckBox

```
android:id="@+id/btnc2"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:text="Dancing"
app:layout_constraintBottom_toBottomOf="parent"
app:layout_constraintEnd_toEndOf="parent"
app:layout_constraintHorizontal_bias="0.23"
app:layout_constraintStart_toStartOf="parent"
app:layout_constraintTop_toTopOf="parent"
app:layout_constraintVertical_bias="0.423" />
```

<CheckBox


```
android:id="@+id/btnc3"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:text="Playing"
app:layout_constraintBottom_toBottomOf="parent"
app:layout_constraintEnd_toEndOf="parent"
app:layout_constraintHorizontal_bias="0.23"
app:layout_constraintStart_toStartOf="parent"
app:layout_constraintTop_toTopOf="parent"
app:layout_constraintVertical_bias="0.519" />
```

<Button

```
android:id="@+id/btns"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:text="SHOW"
app:layout_constraintBottom_toBottomOf="parent"
app:layout_constraintEnd_toEndOf="parent"
app:layout_constraintHorizontal_bias="0.498"
app:layout_constraintStart_toStartOf="parent"
app:layout_constraintTop_toTopOf="parent"
app:layout_constraintVertical_bias="0.67"
android:onClick="display" />
```

<TextView

```
android:id="@+id/txt1"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:text="Hobbies"
app:layout_constraintBottom_toBottomOf="parent"
app:layout_constraintEnd_toEndOf="parent"
app:layout_constraintHorizontal_bias="0.447"
app:layout_constraintStart_toStartOf="parent"
```

```
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_bias="0.082" />
</android.support.constraint.ConstraintLayout>
```

Java Code (MainActivity.java)

```
package com.example.myapplication8;

import android.support.v7.app.AppCompatActivity;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.CheckBox;
import android.widget.Toast;

public class MainActivity extends AppCompatActivity
{
    @Override
    protected void onCreate(Bundle savedInstanceState)
    {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }
    public void display(View v)
    {
        CheckBox c1 = findViewById(R.id.btnc);
        CheckBox c2 = findViewById(R.id.btnc1);
        CheckBox c3 = findViewById(R.id.btnc2);
        CheckBox c4 = findViewById(R.id.btnc3);
        Button b = findViewById(R.id.btns);
        String s = "You are selected:\n";
        if (c1.isChecked())
        {
            s = s + c1.getText().toString();
        }
        if (c2.isChecked())
        {
            s = s + c2.getText().toString();
        }
        if (c3.isChecked())
```

```

    {
        s = s + c3.getText().toString();
    }
    if (c4.isChecked())
    {
        s = s + c4.getText().toString();
    }
    Toast.makeText(this, s, Toast.LENGTH_LONG).show();
}
}

```

Android Project: "Menu Selection App"

Overview:

This application is designed to display a simple food menu that includes common Indian meal items: **Roti**, **Dal**, **Sabji**, and **Rice**. It provides users with a clean interface to view or select these items, making it easy for them to understand available meal options.

Features:

- Displays a list of basic food items.
- Simple and user-friendly layout.
- Organized text elements, making the app easy to read.

XML Layout Code (activity_main.xml)

```

<?xml version="1.0" encoding="utf-8"?>
<android.support.constraint.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <TextView
        android:id="@+id/txt1"

```

```
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:text="MENU"
app:layout_constraintBottom_toBottomOf="parent"
app:layout_constraintEnd_toEndOf="parent"
app:layout_constraintHorizontal_bias="0.473"
app:layout_constraintStart_toStartOf="parent"
app:layout_constraintTop_toTopOf="parent"
app:layout_constraintVertical_bias="0.056" />
```

<CheckBox

```
android:id="@+id/btnc1"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:text="Roti"
app:layout_constraintBottom_toBottomOf="parent"
app:layout_constraintEnd_toEndOf="parent"
app:layout_constraintHorizontal_bias="0.16"
app:layout_constraintStart_toStartOf="parent"
app:layout_constraintTop_toTopOf="parent"
app:layout_constraintVertical_bias="0.19" />
```

<CheckBox

```
android:id="@+id/btnc2"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:text="Daḷ"
app:layout_constraintBottom_toBottomOf="parent"
app:layout_constraintEnd_toEndOf="parent"
app:layout_constraintHorizontal_bias="0.817"
app:layout_constraintStart_toStartOf="parent"
app:layout_constraintTop_toTopOf="parent"
app:layout_constraintVertical_bias="0.19" />
```

<CheckBox

```
android:id="@+id/btnc3"  
android:layout_width="wrap_content"  
android:layout_height="wrap_content"  
android:text="Sabji"  
app:layout_constraintBottom_toBottomOf="parent"  
app:layout_constraintEnd_toEndOf="parent"  
app:layout_constraintHorizontal_bias="0.16"  
app:layout_constraintStart_toStartOf="parent"  
app:layout_constraintTop_toTopOf="parent"  
app:layout_constraintVertical_bias="0.314" />
```

<CheckBox

```
android:id="@+id/btnc4"  
android:layout_width="wrap_content"  
android:layout_height="wrap_content"  
android:text="Rice"  
app:layout_constraintBottom_toBottomOf="parent"  
app:layout_constraintEnd_toEndOf="parent"  
app:layout_constraintHorizontal_bias="0.817"  
app:layout_constraintStart_toStartOf="parent"  
app:layout_constraintTop_toTopOf="parent"  
app:layout_constraintVertical_bias="0.314" />
```

<Button

```
android:id="@+id/btno"  
android:layout_width="wrap_content"  
android:layout_height="wrap_content"  
android:text="Order Now"  
app:layout_constraintBottom_toBottomOf="parent"  
app:layout_constraintEnd_toEndOf="parent"  
app:layout_constraintHorizontal_bias="0.498"
```

```
app:layout_constraintStart_toStartOf="parent"
app:layout_constraintTop_toTopOf="parent"
app:layout_constraintVertical_bias="0.424"
android:onClick="menu" />
```

<TextView

```
    android:id="@+id/txt2"
    android:layout_width="147dp"
    android:layout_height="97dp"
    android:text="Order Finalized:"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.492"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintVertical_bias="0.669" />
```

</android.support.constraint.ConstraintLayout>

Java Code (MainActivity.java)

```
package com.example.myapplication9;

import android.support.v7.app.AppCompatActivity;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.CheckBox;
import android.widget.EditText;
import android.widget.TextView;

public class MainActivity extends AppCompatActivity {

    @Override
```

```
protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_main);
}

public void menu(View v){
    CheckBox c1 = findViewById(R.id.btnc1);
    CheckBox c2 = findViewById(R.id.btnc2);
    CheckBox c3 = findViewById(R.id.btnc3);
    CheckBox c4 = findViewById(R.id.btnc4);

    Button b = findViewById(R.id.btno);
    TextView t1 = findViewById(R.id.txt2);
    b.setOnClickListener(V → {
        StringBuilder Order = new StringBuilder();

        boolean isSelected = false;

        if (c1.isChecked()){
            Order.append("Roti\n");
            isSelected = true;
        }
        if (c2.isChecked()){
            Order.append("Dal\n");
            isSelected = true;
        }
        if (c3.isChecked()){
            Order.append("Sabji\n");
            isSelected = true;
        }
        if (c4.isChecked()){
            Order.append("Rice\n");
            isSelected = true;
        }
    }
}
```

```

        if (!isSelected){
            t1.setText("You have not selected any item.");
        }else {
            t1.setText("You have selected:\n" +
Order.toString());
        }
    });
}
}
}

```

Day : 4 Android Project: "Gender Selector App"

Date: 06/06/2025

Overview:

This is a simple Android application that allows users to select their gender by choosing either **Male** or **Female**. The app presents two options on the screen in the form of buttons (or radio buttons), ensuring that the user can select one at a time. This type of app is commonly used in forms, registration pages, and surveys where gender information is required.

Functionality:

- The app displays two options: **Male** and **Female**.
- The user can select one gender option by clicking/tapping on the respective button.
- The app ensures only one option is active at a time, helping avoid multiple
- selections (if RadioButton or similar is used).
- It can be extended to display the selected gender or proceed to another activity.

XML Layout Code (activity_main.xml)

```

<?xml version="1.0" encoding="utf-8"?>
<android.support.constraint.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"

```



```
tools:context=".MainActivity">
```

```
<TextView
```

```
    android:id="@+id/txt1"  
    android:layout_width="wrap_content"  
    android:layout_height="wrap_content"  
    android:text="GENDER:"  
    app:layout_constraintBottom_toBottomOf="parent"  
    app:layout_constraintEnd_toEndOf="parent"  
    app:layout_constraintHorizontal_bias="0.45"  
    app:layout_constraintStart_toStartOf="parent"  
    app:layout_constraintTop_toTopOf="parent"  
    app:layout_constraintVertical_bias="0.125" />
```

```
<Button
```

```
    android:id="@+id/btn3"  
    android:layout_width="wrap_content"  
    android:layout_height="wrap_content"  
    android:text="SHOW"  
    app:layout_constraintBottom_toBottomOf="parent"  
    app:layout_constraintEnd_toEndOf="parent"  
    app:layout_constraintHorizontal_bias="0.498"  
    app:layout_constraintStart_toStartOf="parent"  
    app:layout_constraintTop_toTopOf="parent"  
    app:layout_constraintVertical_bias="0.4"  
    android:onClick="show" />
```

```
<RadioGroup
```

```
    android:id="@+id/btn1"  
    android:layout_width="158dp"  
    android:layout_height="116dp"  
    app:layout_constraintBottom_toBottomOf="parent"  
    app:layout_constraintEnd_toEndOf="parent"
```

```
app:layout_constraintHorizontal_bias="0.595"
app:layout_constraintStart_toStartOf="parent"
app:layout_constraintTop_toTopOf="parent"
app:layout_constraintVertical_bias="0.196">
```

```
<RadioButton
```

```
    android:id="@+id/btn2"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Male" />
```

```
<RadioButton
```

```
    android:id="@+id/btn4"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Female" />
```

```
</RadioGroup>
```

```
</android.support.constraint.ConstraintLayout>
```

Java Code (MainActivity.java)

```
package com.example.myapplication10;

import android.support.v7.app.AppCompatActivity;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.RadioButton;
import android.widget.RadioGroup;
import android.widget.Toast;

public class MainActivity extends AppCompatActivity {
```

```

@Override
protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_main);
}
/* public void show(View v){
    RadioButton r = findViewById(R.id.btn2);
    RadioButton r1 = findViewById(R.id.btn4);
    Button b = findViewById(R.id.btn3);
    RadioGroup r2 = findViewById(R.id.btn1);

    String s = "You are Selected:\n";

    if(r.isChecked()){
        s = s + r.getText().toString();
    }
    if(r1.isChecked()){
        s = s + r1.getText().toString();
    }
    Toast.makeText(this,s,Toast.LENGTH_LONG).show();

}*/
public void display(View v){
    RadioGroup r = findViewById(R.id.btn1);
    int Value = r.getCheckedRadioButtonId();
    RadioButton r1 = findViewById(Value);
    String gender = r1.getText().toString();
    Toast.makeText(this,gender,Toast.LENGTH_LONG).show();
}
}

```

Day : 5 Android Project: "Toggle Switch App"

Date: 07/06/2025

Overview:

This is a basic Android app that demonstrates the use of a Toggle Switch in a mobile interface. The app allows users to switch between two states: ON and OFF, with the switch changing its position accordingly. Toggle switches are commonly used in settings screens or any app where users need to enable or disable a feature with a single tap.

The app provides a simple, minimalistic interface that clearly shows the current state of the toggle.

Functionality:

- The app displays a **Toggle Switch** on the screen.
- The user can tap the toggle to change its state between **ON** and **OFF**.
- The app reflects the current state visually.
- The state can be used to trigger actions like turning features on or off (e.g., sound, WiFi, notifications).

XML Layout Code (activity_main.xml)

```
<?xml version="1.0" encoding="utf-8"?>
<android.support.constraint.ConstraintLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <ToggleButton
        android:id="@+id/btn1"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="ToggleButton"
        app:layout_constraintBottom_toBottomOf="parent"
```

```
app:layout_constraintEnd_toEndOf="parent"
app:layout_constraintStart_toStartOf="parent"
app:layout_constraintTop_toTopOf="parent"
android:onClick="Tog" />
```

<TextView

```
    android:id="@+id/txt1"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text=""
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.498"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintVertical_bias="0.106" />
```

</android.support.constraint.ConstraintLayout>

Java Code (MainActivity.java)

```
package com.example.myapplication11;

import android.support.v7.app.AppCompatActivity;
import android.os.Bundle;
import android.view.View;
import android.widget.TextView;
import android.widget.ToggleButton;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }
}
```

```

    }
    public void Tog(View v){
        ToggleButton t = findViewById(R.id.btn1);
        TextView t1 = findViewById(R.id.txt1);
        //t1.setText(t.getText().toString());
        if(t.isChecked()){
            t1.setText("Toggle is ON.");
        } else {
            t1.setText("Toggle is OFF.");
        }
    }
}

```

Day : 6 Android Project: "Simple Multi-Activity Navigation App"

Date: 09/06/2025

Overview:

This is a simple Android app designed to demonstrate **multi-activity navigation**. The app consists of multiple activities (screens) where the user moves from one screen to the next by clicking a button. This app helps beginners understand how to create, connect, and navigate between different activities in Android.

Functionality:

- The app starts with the **MainActivity** that shows a button.
- When the button is clicked, it opens the **SecondActivity**.
- The **SecondActivity** also displays a button.
- When this second button is clicked, it opens the **ThirdActivity** (or a blank next activity).

The user can see the change of screen each time they click a button, showing proper navigation.

XML Layout Code (activity_main.xml)

```

<?xml version="1.0" encoding="utf-8"?>

<android.support.constraint.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <Button
        android:id="@+id/btn1"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Button"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.498"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_bias="0.166"
        android:onClick="demo" />
</android.support.constraint.ConstraintLayout>

```

Java Code (MainActivity.java)

```

package com.example.myapplication12;

import android.content.Intent;
import android.support.v7.app.AppCompatActivity;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.Toast;

```

```

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }

    public void demo(View v){
        Button b = findViewById(R.id.btn1);
        String s = "Hello";
        Intent i = new Intent(this, MainActivity2.class);
        startActivity(i);
        // Toast.makeText(this,s,Toast.LENGTH_LONG).show();
    }
}

```

XML Layout Code (activity_main2.xml)

```

<?xml version="1.0" encoding="utf-8"?>

<android.support.constraint.ConstraintLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity2">

    <Button
        android:id="@+id/btn2"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Button"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"

```



```
        app:layout_constraintHorizontal_bias="0.498"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_bias="0.174"
        android:onClick="show" />
</android.support.constraint.ConstraintLayout>
```

Java Code (MainActivity2.java)

```
package com.example.myapplication12;

import android.content.Intent;
import android.support.v7.app.AppCompatActivity;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.Toast;

public class MainActivity2 extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main2);
    }

    public void show(View v){
        Button b1 = findViewById(R.id.btn2);
        String s1 = "Hi";
        Intent i = new Intent(this, MainActivity3.class);
        startActivity(i);
        //Toast.makeText(this,s1,Toast.LENGTH_LONG).show();
    }
}
```

XML Layout Code (activity_main3.xml)

```
<?xml version="1.0" encoding="utf-8"?>

<android.support.constraint.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity3">
</android.support.constraint.ConstraintLayout>
```

Java Code (MainActivity3.java)

```
package com.example.myapplication12;

import android.support.v7.app.AppCompatActivity;
import android.os.Bundle;

public class MainActivity3 extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main3);
    }
}
```

Week : 3

Day : 1 Android Project: "Android Application Demonstrating Button Events and Web Redirection"

Date: 10/06/2025

Overview:

This is a basic Android app that showcases button event handling and web redirection using an ImageButton. The app is designed for beginners who want to understand how button clicks can trigger actions and how to open external web pages (like Google) from an Android app.

The app consists of:

- One ImageButton (showing a Google logo or icon).
- When the button is clicked, it redirects the user to the Google website or Google app.

This demonstrates how Android apps can interact with web browsers or external apps through intents.

Functionality:

- The app launches with a screen containing an ImageButton.
- The ImageButton is linked to an event listener that detects clicks.
- When the user clicks the button, an Intent is created to open the Google page .

XML Layout Code (activity_main.xml)

```
<?xml version="1.0" encoding="utf-8"?>

<android.support.constraint.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <Button
        android:id="@+id/btn1"
        android:layout_width="wrap_content"
```

```
android:layout_height="wrap_content"
android:text="Click here"
app:layout_constraintBottom_toBottomOf="parent"
app:layout_constraintEnd_toEndOf="parent"
app:layout_constraintHorizontal_bias="0.467"
app:layout_constraintStart_toStartOf="parent"
app:layout_constraintTop_toTopOf="parent"
app:layout_constraintVertical_bias="0.232"
android:onClick="show1" />
```

<Button

```
android:id="@+id/btn2"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:text="GO"
app:layout_constraintBottom_toBottomOf="parent"
app:layout_constraintEnd_toEndOf="parent"
app:layout_constraintHorizontal_bias="0.467"
app:layout_constraintStart_toStartOf="parent"
app:layout_constraintTop_toTopOf="parent"
app:layout_constraintVertical_bias="0.468"
android:onClick="demo" />
```

</android.support.constraint.ConstraintLayout>

Java Code (MainActivity.java)

```
package com.example.myapplication13;

import android.content.Intent;
import android.support.v7.app.AppCompatActivity;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.Toast;
```

```

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }
    public void show1(View v) {
        Button b1 = findViewById(R.id.btn1);
        String s = "Hello";
        Toast.makeText(this, s, Toast.LENGTH_LONG).show();
    }
    public void demo(View v){
        Button b2 = findViewById(R.id.btn2);
        Intent i = new Intent(this, MainActivity2.class);
        startActivity(i);
    }
}

```

XML Layout Code (activity_main2.xml)

```

<?xml version="1.0" encoding="utf-8"?>

<android.support.constraint.ConstraintLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity2">

    <ImageButton
        android:id="@+id/btn3"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"

```

```
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:srcCompat="@mipmap/google1"
        android:onClick="show2"/>
</android.support.constraint.ConstraintLayout>
```

Java Code (MainActivity2.java)

```
package com.example.myapplication13;

import android.content.Intent;
import android.net.Uri;
import android.support.v7.app.AppCompatActivity;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;

public class MainActivity2 extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main2);
    }

    public void show2(View v){

        Intent i= new Intent(android.content.Intent.ACTION_VIEW,
Uri.parse("http://www.google.co.in"));
        startActivity(i);
    }
}
```

Day 2: Android Project: "Sum Calculator with Result Display"

Date: 11/06/2025

Overview:

This is a basic **Android calculator app** that allows the user to input two numbers and calculate their sum. The result is **displayed on a second activity**. This app helps beginners understand how to **pass data between activities** using **Intents** in Android.

The app contains:

- Two **EditText** fields to enter Number 1 and Number 2.
- A **Button** labeled **Sum** to trigger the calculation.
- A **TextView** on the second activity to display the result.

Functionality:

- The user enters two numbers in the input fields.
- When the **Sum** button is clicked:
 - The app **retrieves the input values** from the EditText fields.
 - It **calculates the sum** of these two numbers.
 - The app **creates an Intent** to start a new activity.
 - The sum is **passed to the next activity** using the Intent.
 - The second activity receives the sum and displays it clearly in a TextView (e.g., "Sum is: 30").

XML Layout Code (activity_main.xml)

```
<?xml version="1.0" encoding="utf-8"?>

<android.support.constraint.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
```

```
tools:context=".MainActivity">
```

```
<TextView
```

```
    android:id="@+id/txt1"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Number1:"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.235"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintVertical_bias="0.176" />
```

```
<TextView
```

```
    android:id="@+id/txt2"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Number2:"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.235"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintVertical_bias="0.303" />
```

```
<EditText
```

```
    android:id="@+id/pl1"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:ems="10"
    android:inputType="textPersonName"
    android:text=""
```



```
app:layout_constraintBottom_toBottomOf="parent"
app:layout_constraintEnd_toEndOf="parent"
app:layout_constraintHorizontal_bias="0.786"
app:layout_constraintStart_toStartOf="parent"
app:layout_constraintTop_toTopOf="parent"
app:layout_constraintVertical_bias="0.144" />
```

<EditText

```
android:id="@+id/plt2"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:ems="10"
android:inputType="textPersonName"
android:text=""
app:layout_constraintBottom_toBottomOf="parent"
app:layout_constraintEnd_toEndOf="parent"
app:layout_constraintHorizontal_bias="0.791"
app:layout_constraintStart_toStartOf="parent"
app:layout_constraintTop_toTopOf="parent"
app:layout_constraintVertical_bias="0.276" />
```

<Button

```
android:id="@+id/btns"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:text="SUM"
app:layout_constraintBottom_toBottomOf="parent"
app:layout_constraintEnd_toEndOf="parent"
app:layout_constraintStart_toStartOf="parent"
app:layout_constraintTop_toTopOf="parent"
android:onClick="sum" />
```

</android.support.constraint.ConstraintLayout>

Java Code (MainActivity.java)

```
package com.example.myapplication14;

import android.content.Intent;
import android.support.v7.app.AppCompatActivity;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }

    public void sum(View v){
        TextView t1 = findViewById(R.id.txt1);
        TextView t2 = findViewById(R.id.txt2);
        EditText e1 = findViewById(R.id.plt1);
        EditText e2 = findViewById(R.id.plt2);
        Button b = findViewById(R.id.btns);

        String s1 = e1.getText().toString();
        String s2 = e2.getText().toString();

        Double result = Double.parseDouble(s1) +
        Double.parseDouble(s2);

        Intent i = new Intent(this, MainActivity2.class);
        i.putExtra("sum", result);
        startActivity(i);
    }
}
```

```
}  
}
```

XML Layout Code (activity_main2.xml)

```
<?xml version="1.0" encoding="utf-8"?>  
  
<android.support.constraint.ConstraintLayout  
    xmlns:android="http://schemas.android.com/apk/res/android"  
        xmlns:app="http://schemas.android.com/apk/res-auto"  
        xmlns:tools="http://schemas.android.com/tools"  
        android:layout_width="match_parent"  
        android:layout_height="match_parent"  
        tools:context=".MainActivity2">  
  
    <TextView  
        android:id="@+id/txt3"  
        android:layout_width="136dp"  
        android:layout_height="37dp"  
        android:text=""  
        app:layout_constraintBottom_toBottomOf="parent"  
        app:layout_constraintEnd_toEndOf="parent"  
        app:layout_constraintHorizontal_bias="0.467"  
        app:layout_constraintStart_toStartOf="parent"  
        app:layout_constraintTop_toTopOf="parent"  
        app:layout_constraintVertical_bias="0.127" />  
</android.support.constraint.ConstraintLayout>
```

Java Code (MainActivity2.java)

```
package com.example.myapplication14;  
  
import android.content.Intent;  
import android.support.v7.app.AppCompatActivity;  
import android.os.Bundle;  
import android.view.View;  
import android.widget.EditText;
```

```
import android.widget.TextView;

public class MainActivity2 extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main2);
        TextView t1 = findViewById(R.id.txt3);
        Double sum = getIntent().getDoubleExtra("sum", 0.0);
        t1.setText("Sum is: "+sum);
    }
}
```

Day 3: Android Project:

"TextView with Different Greetings and Colors" OR "Button Widgets with Different Labels and Colors"

Date: 12/06/2025

Overview:

This application displays various **greetings** like Hello, Hi, Good Morning, etc., using **TextView widgets** in Android. Each TextView is styled with a **different color**, helping users understand how to modify text appearance in an Android app.

Functionality:

- The app displays multiple TextViews on the screen.
- Each TextView contains a **different greeting message**.
- The color of each greeting is **customized** (e.g., one is red, another blue, another green).
- The app demonstrates how to use the **android:textColor** property in XML or set color using Java code.

XML Layout Code (activity_main.xml)

```
<?xml version="1.0" encoding="utf-8"?>
```

```
<android.support.constraint.ConstraintLayout
```

```
xmlns:android="http://schemas.android.com/apk/res/android"
```

```
xmlns:app="http://schemas.android.com/apk/res-auto"
```

```
xmlns:tools="http://schemas.android.com/tools"
```

```
android:layout_width="match_parent"
```

```
android:layout_height="match_parent"
```

```
tools:context=".MainActivity">
```

```
<TextView
```

```
    android:id="@+id/txt1"
```

```
    android:layout_width="wrap_content"
```

```
    android:layout_height="wrap_content"
```

```
    android:text="@string/S1"
```

```
    android:textColor="#3F51B5"
```

```
    app:layout_constraintBottom_toBottomOf="parent"
```

```
    app:layout_constraintEnd_toEndOf="parent"
```

```
    app:layout_constraintHorizontal_bias="0.501"
```

```
    app:layout_constraintStart_toStartOf="parent"
```

```
    app:layout_constraintTop_toTopOf="parent"
```

```
    app:layout_constraintVertical_bias="0.237" />
```

```
<TextView
```

```
    android:id="@+id/txt2"
```

```
    android:layout_width="wrap_content"
```

```
    android:layout_height="wrap_content"
```

```
    android:text="@string/S2"
```

```
    android:textColor="#4CAF50"
```

```
    app:layout_constraintBottom_toBottomOf="parent"
```

```
    app:layout_constraintEnd_toEndOf="parent"
```

```
    app:layout_constraintHorizontal_bias="0.501"
```

```
    app:layout_constraintStart_toStartOf="parent"
```

```
app:layout_constraintTop_toTopOf="parent"
app:layout_constraintVertical_bias="0.32" />
```

<TextView

```
    android:id="@+id/txt3"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="@string/S3"
    android:textColor="#F44336"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.501"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintVertical_bias="0.428" />
```

<TextView

```
    android:id="@+id/txt4"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="@string/S4"
    android:textColor="#CDDC39"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent" />
```

</android.support.constraint.ConstraintLayout>

Java Code (MainActivity.java)

```
package com.example.myapplication15;
```

```
import android.support.v7.app.AppCompatActivity;
```

```

import android.os.Bundle;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }
}

```

Strings.xml

```

<resources>
    <string name="app_name">My Application15</string>
    <string name="S1">HELLO</string>
    <string name="S2">HI</string>
    <string name="S3">GOOD MORNING</string>
    <string name="S4">NAMASTE</string>
</resources>

```

Colors.xml

```

<?xml version="1.0" encoding="utf-8"?>
<resources>
    <color name="purple_200">#FFBB86FC</color>
    <color name="purple_500">#FF6200EE</color>
    <color name="purple_700">#FF3700B3</color>
    <color name="teal_200">#FF03DAC5</color>
    <color name="teal_700">#FF018786</color>
    <color name="black">#FF000000</color>
    <color name="white">#FFFFFFFF</color>
    <color name="red">#F44336</color>
    <color name="blue">#3F51B5</color>
    <color name="Yellow">#CDDC39</color>
    <color name="green">#4CAF50</color>

```

</resources>

Android Project: "Button Widgets with Different Labels and Colors"

Overview:

This Android app demonstrates the use of **multiple Button widgets**, each labeled with a different action like **Click**, **Click Here**, **Shop**, and **Shop Now**. Every button is styled with **unique colors** to make the interface more vibrant and user-friendly.

The purpose of this app is to help users understand how to add and style multiple buttons in an Android layout.

Functionality:

The app displays **4 buttons** on the screen:

- **Click**
- **Click Here**
- **Shop**
- **Shop Now**

Each button has:

- A **different label** (text)
- A **different background color** (for easy identification)

XML Layout Code (activity_main.xml)

```
<?xml version="1.0" encoding="utf-8"?>
```

```
<android.support.constraint.ConstraintLayout  
xmlns:android="http://schemas.android.com/apk/res/android"  
    xmlns:app="http://schemas.android.com/apk/res-auto"  
    xmlns:tools="http://schemas.android.com/tools"  
    android:layout_width="match_parent"  
    android:layout_height="match_parent"  
    tools:context=".MainActivity">
```

```
<Button
```



```
android:id="@+id/btn1"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:text="@string/s1"
android:background="@color/green"
app:layout_constraintBottom_toBottomOf="parent"
app:layout_constraintEnd_toEndOf="parent"
app:layout_constraintHorizontal_bias="0.476"
app:layout_constraintStart_toStartOf="parent"
app:layout_constraintTop_toTopOf="parent"
app:layout_constraintVertical_bias="0.265" />
```

<Button

```
android:id="@+id/btn2"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:text="@string/s2"
android:background="@color/pink"
app:layout_constraintBottom_toBottomOf="parent"
app:layout_constraintEnd_toEndOf="parent"
app:layout_constraintHorizontal_bias="0.476"
app:layout_constraintStart_toStartOf="parent"
app:layout_constraintTop_toTopOf="parent"
app:layout_constraintVertical_bias="0.415" />
```

<Button

```
android:id="@+id/btn3"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:text="@string/s3"
android:background="@color/blue"
app:layout_constraintBottom_toBottomOf="parent"
app:layout_constraintEnd_toEndOf="parent"
```

```
app:layout_constraintHorizontal_bias="0.476"
app:layout_constraintStart_toStartOf="parent"
app:layout_constraintTop_toTopOf="parent"
app:layout_constraintVertical_bias="0.572" />
```

<Button

```
android:id="@+id/btn4"
android:layout_width="89dp"
android:layout_height="45dp"
android:text="@string/s4"
android:background="@color/purple"
app:layout_constraintBottom_toBottomOf="parent"
app:layout_constraintEnd_toEndOf="parent"
app:layout_constraintHorizontal_bias="0.478"
app:layout_constraintStart_toStartOf="parent"
app:layout_constraintTop_toTopOf="parent"
app:layout_constraintVertical_bias="0.72" />
```

</android.support.constraint.ConstraintLayout>

Java Code (MainActivity.java)

```
package com.example.myapplication16;

import android.support.v7.app.AppCompatActivity;
import android.os.Bundle;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }
}
```

Strings.xml

```
<resources>
    <string name="app_name">My Application16</string>
    <string name="s1">Click here</string>
    <string name="s2">Shop</string>
    <string name="s3">Shop Now</string>
    <string name="s4">Click</string>
</resources>
```

Colors.xml

```
<?xml version="1.0" encoding="utf-8"?>
<resources>
    <color name="purple_200">#FFBB86FC</color>
    <color name="purple_500">#FF6200EE</color>
    <color name="purple_700">#FF3700B3</color>
    <color name="teal_200">#FF03DAC5</color>
    <color name="teal_700">#FF018786</color>
    <color name="black">#FF000000</color>
    <color name="white">#FFFFFFFF</color>
    <color name="purple">#5C2EAC</color>
    <color name="blue">#2A6885</color>
    <color name="green">#8DE820</color>
    <color name="pink">#CF5E84</color>

</resources>
```

Day 4: Android Project:

"Change Text Size App" OR "ID and Name Submit App"

Date: 13/06/2025

Overview:

This Android app demonstrates how to dynamically **change the text size of a TextView** using buttons. It is designed to help beginners understand how button clicks can be used to change the appearance of text on the screen. The app contains a **TextView** (showing the word **Hello**) and **three buttons** labeled:

- **Large**
- **Medium**
- **Small**

Each button adjusts the text size of **Hello** based on the user's selection.

Functionality:

- The app displays:
 - A **TextView** showing the text: Hello
 - Three **Button widgets**: Large, Medium, and Small
- When the user clicks:
 - **Large button** → The text Hello becomes large in size.
 - **Medium button** → The text Hello changes to medium size.
 - **Small button** → The text Hello changes to small size
- This demonstrates how we can:
 - Use button clicks to control the font size of text.
 - Provide immediate feedback to the user through visual changes.

XML Layout Code (activity_main.xml)

```
<?xml version="1.0" encoding="utf-8"?>
<android.support.constraint.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
```

```
tools:context=".MainActivity">
```

```
<TextView
```

```
    android:id="@+id/txt1"  
    android:layout_width="wrap_content"  
    android:layout_height="wrap_content"  
    android:text="Hello"  
    app:layout_constraintBottom_toBottomOf="parent"  
    app:layout_constraintEnd_toEndOf="parent"  
    app:layout_constraintHorizontal_bias="0.47"  
    app:layout_constraintStart_toStartOf="parent"  
    app:layout_constraintTop_toTopOf="parent"  
    app:layout_constraintVertical_bias="0.134" />
```

```
<Button
```

```
    android:id="@+id/btn1"  
    android:layout_width="114dp"  
    android:layout_height="64dp"  
    android:text="Large"  
    app:layout_constraintBottom_toBottomOf="parent"  
    app:layout_constraintEnd_toEndOf="parent"  
    app:layout_constraintHorizontal_bias="0.476"  
    app:layout_constraintStart_toStartOf="parent"  
    app:layout_constraintTop_toTopOf="parent"  
    app:layout_constraintVertical_bias="0.304"  
    android:onClick="large" />
```

```
<Button
```

```
    android:id="@+id/btn2"  
    android:layout_width="118dp"  
    android:layout_height="67dp"  
    android:text="Medium"  
    app:layout_constraintBottom_toBottomOf="parent"
```

```
app:layout_constraintEnd_toEndOf="parent"
app:layout_constraintHorizontal_bias="0.484"
app:layout_constraintStart_toStartOf="parent"
app:layout_constraintTop_toTopOf="parent"
app:layout_constraintVertical_bias="0.469"
android:onClick="medium" />
```

<Button

```
android:id="@+id/btn3"
android:layout_width="129dp"
android:layout_height="70dp"
android:text="Small"
app:layout_constraintBottom_toBottomOf="parent"
app:layout_constraintEnd_toEndOf="parent"
app:layout_constraintHorizontal_bias="0.503"
app:layout_constraintStart_toStartOf="parent"
app:layout_constraintTop_toTopOf="parent"
app:layout_constraintVertical_bias="0.661"
android:onClick="small" />
```

</android.support.constraint.ConstraintLayout>

Java Code (MainActivity.java)

```
package com.example.myapplication17;

import android.support.v7.app.AppCompatActivity;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.TextView;

public class MainActivity extends AppCompatActivity {
    TextView t1;
```

```

@Override
protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_main);
    t1= findViewById(R.id.txt1);
}
public void large(View v){
    Button b1 = findViewById(R.id.btn1);
    t1.setTextSize(100);
}
public void medium(View v){
    Button b2 = findViewById(R.id.btn2);
    t1.setTextSize(50);
}
public void small(View v){
    Button b3 = findViewById(R.id.btn3);
    t1.setTextSize(25);
}
}

```

Android Project: "ID and Name Submit App"

Overview:

This Android app is designed to **collect basic user input** (ID and Name) through two input fields and display this information on another screen (activity) after the user clicks the **Submit** button. It demonstrates the use of **EditText for input**, **Button for action**, and **Intents for activity navigation** while passing data between activities.

Functionality:

- The app displays on the first activity:
 - An **EditText** for the user to enter their **ID**
 - An **EditText** for the user to enter their **Name**

- A **Submit button**
- What happens:
 - The user enters their ID and Name in the respective fields.
 - When the **Submit button** is clicked, the app:
 - **Sends the entered ID and Name** to the next activity using an **Intent**
 - **Navigates to the next activity (screen)**
 - Displays the entered ID and Name clearly in a **TextView** or similar widget.
 - **XML Layout Code (activity_main.xml)**

```
<?xml version="1.0" encoding="utf-8"?>
```

```
<android.support.constraint.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:app="http://schemas.android.com/apk/res-auto"
xmlns:tools="http://schemas.android.com/tools"
android:layout_width="match_parent"
android:layout_height="match_parent"
tools:context=".MainActivity">
```

```
<TextView
```

```
    android:id="@+id/txt1"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="ID"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.133"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintVertical_bias="0.171" />
```

```
<TextView
```



```
android:id="@+id/txt2"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:text="Name"
app:layout_constraintBottom_toBottomOf="parent"
app:layout_constraintEnd_toEndOf="parent"
app:layout_constraintHorizontal_bias="0.133"
app:layout_constraintStart_toStartOf="parent"
app:layout_constraintTop_toTopOf="parent"
app:layout_constraintVertical_bias="0.285" />
```

<EditText

```
android:id="@+id/plt1"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:ems="10"
android:inputType="textPersonName"
android:text=""
app:layout_constraintBottom_toBottomOf="parent"
app:layout_constraintEnd_toEndOf="parent"
app:layout_constraintHorizontal_bias="0.671"
app:layout_constraintStart_toStartOf="parent"
app:layout_constraintTop_toTopOf="parent"
app:layout_constraintVertical_bias="0.139" />
```

<EditText

```
android:id="@+id/plt2"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:ems="10"
android:inputType="textPersonName"
android:text=""
app:layout_constraintBottom_toBottomOf="parent"
```

```
app:layout_constraintEnd_toEndOf="parent"
app:layout_constraintHorizontal_bias="0.676"
app:layout_constraintStart_toStartOf="parent"
app:layout_constraintTop_toTopOf="parent"
app:layout_constraintVertical_bias="0.258" />
```

<Button

```
android:id="@+id/btn1"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:text="Submit"
app:layout_constraintBottom_toBottomOf="parent"
app:layout_constraintEnd_toEndOf="parent"
app:layout_constraintStart_toStartOf="parent"
app:layout_constraintTop_toTopOf="parent"
android:onClick="demo" />
```

</android.support.constraint.ConstraintLayout>

Java Code (MainActivity.java)

```
package com.example.myapplication18;

import android.content.Intent;
import android.support.v7.app.AppCompatActivity;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }
}
```

```

    }
    public void demo(View v){
        EditText e1 = findViewById(R.id.plt1);
        EditText e2 = findViewById(R.id.plt2);
        Button b1 = findViewById(R.id.btn1);

        String s1 = e1.getText().toString();
        String s2 = e2.getText().toString();
        Intent i = new Intent(this,MainActivity2.class);
        i.putExtra("Id",s1);
        i.putExtra("Name",s2);
        startActivity(i);
    }
}

```

XML Layout Code (activity_main2.xml)

```

<?xml version="1.0" encoding="utf-8"?>
<android.support.constraint.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity2">

    <EditText
        android:id="@+id/plt3"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:ems="10"
        android:inputType="textPersonName"
        android:text=""
        app:layout_constraintBottom_toBottomOf="parent"

```

```
app:layout_constraintEnd_toEndOf="parent"
app:layout_constraintHorizontal_bias="0.786"
app:layout_constraintStart_toStartOf="parent"
app:layout_constraintTop_toTopOf="parent"
app:layout_constraintVertical_bias="0.118" />
```

<EditText

```
android:id="@+id/plt4"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:ems="10"
android:inputType="textPersonName"
android:text=""
app:layout_constraintBottom_toBottomOf="parent"
app:layout_constraintEnd_toEndOf="parent"
app:layout_constraintHorizontal_bias="0.791"
app:layout_constraintStart_toStartOf="parent"
app:layout_constraintTop_toTopOf="parent"
app:layout_constraintVertical_bias="0.284" />
```

<TextView

```
android:id="@+id/txt3"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:text="ID"
app:layout_constraintBottom_toBottomOf="parent"
app:layout_constraintEnd_toEndOf="parent"
app:layout_constraintHorizontal_bias="0.127"
app:layout_constraintStart_toStartOf="parent"
app:layout_constraintTop_toTopOf="parent"
app:layout_constraintVertical_bias="0.15" />
```

<TextView

```

        android:id="@+id/txt4"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Name"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.127"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_bias="0.311" />
</android.support.constraint.ConstraintLayout>

```

Java Code (MainActivity2.java)

```

package com.example.myapplication18;

import android.support.v7.app.AppCompatActivity;
import android.os.Bundle;
import android.widget.EditText;

public class MainActivity2 extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main2);
        EditText e3 = findViewById(R.id.plt3);
        EditText e4 = findViewById(R.id.plt4);
        String s3 = getIntent().getStringExtra("Id");
        String s4 = getIntent().getStringExtra("Name");
        e3.setText(s3);
        e4.setText(s4);
    }
}

```

